

Graph: Find Center Node

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OBSERVATION

- Center node has degree = $n - 1$.
- Every other node has degree = 1 .

ALGORITHM

- Create degree array.
- Traverse all edges and increment degree of both endpoints.
- Return the node whose degree is $n - 1$.

COMPLEXITY

Time Complexity: $O(n)$

Space Complexity: $O(n)$

IMPLEMENTATION CODE

```
class Solution {
public:
    int findCenter(vector<vector<int>>& edges) {
        //basic idea is degree counting
        //star node ki degree==n-1
        //baaki sab ki degree==1

        int n=edges.size()+1;
        vector<int> degree(n+1,0); //+1 kr rhe cause 1 to n ki
        label he

        for(auto &it:edges){

            int u=it[0];
            int v=it[1];

            degree[u]++;
            degree[v]++;
        }

        for(int i=1;i<=n;i++){

            if(degree[i]==n-1) return i;

        }

        return -1;
    }
};
```